Discrete

→ PDNF (SOP)

→ PCNF (POS)

Implication $P \to q$

Converse $q \to P$

Inverse $\neg P \to \neg q$

Contrapositive $\neg q \to \neg P$ $[\neg q \to \neg P]$

$P \to q$.

1. if P then q
2. P is sufficient Condition for q
3. q when P
4. a necessary Condition for P is q
5. q unless $\neg P$
6. P only if q
7. q follows from P

1. P iff q $(P \leftrightarrow q)$
2. if P then q & vice versa
3. P is necessary & sufficient Condition for q.

Rules of Inferences:

1. $\dfrac{P \quad P \to q}{q}$
2. $\dfrac{\neg q \quad P \to q}{\neg P}$
3. $\dfrac{P \to q \quad q \to r}{P \to r}$
4. $\dfrac{\neg P}{P \vee q}$
5. $\dfrac{P}{P \vee q}$
6. $\dfrac{(P \wedge q) \to r}{(P \to q) \to r}$
7. $\dfrac{P \vee q \quad \neg P \vee r}{q \vee r}$

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Set Theory :

→ Symmetric diff / Boolean sum : $(A-B) \cup (B-A)$ or $(A \cup B) - (A \cap B)$

→ Anti-symmetric : if $(x R y \text{ & } y R x) \Rightarrow x = y$. for $x, y \in A$.

→ Equivalence
- → Reflective
- → Symmetric
- → Transitive

→ Total order
- ↓ Partial order
  - → Reflexive
  - Anti → Anti-symmetric
  - → Transitive

No of Equivalence Rel^n with Bell No.

1

1, 2

2, 3, 5

5, 7, 10, 15

15.

Lattices

→ Least upper Bound ($\vee$) → Join → for All pair → Join Semi Lattice

→ Greatest Lower Bound ($\wedge$) → Meet → Meet semi Lattice

Lattice = JSL + MSL → All lattices follows
- ↳ Commutative, Associative, Idempotent, Absorption.

But not distributive.

→ Distributive [Max^m ⊥ Compliment]

→ Bounded [Every finite Lattice Bounded, Compliment applicable only on bounded]

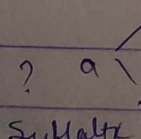
→ Compliment [each have compliment]

→ Boolean [Distributive + Compliment] → Unique Compliment

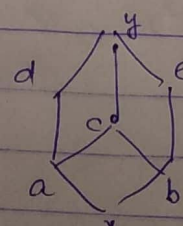
↳ Compliment : $(a \wedge b) = 0$ & $(a \vee b) = I_0$

Sublattice Que-

{x, a, b, y}



? Sublattice ? : LUB(a, b) = y.

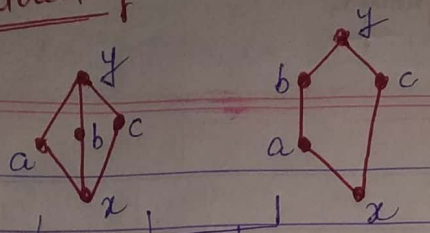


→ LUB(a, b) = c

→ Not equal, so, Not sublattice.

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Sup. & Interesting



Bcz of any (a,b,c)

↳ if these two present as Sub-lattice in any Lattice. (any of them) then that Lattice will not be distributive.

→ A Lattice with 4 or fewer ele is distributive
→ every total order set is distributive
→ Every sublattice of a Lattice is distributive.

Algebra

→ Semi-Group
↳ Associative

→ Monoid
↳ Associative
↳ Identity ele

→ Group
↳ Associative
↳ Identity
↳ Inverse

→ Abelian Group
→ Associative
→ Identity
→ Inverse
→ Commutative

→ finite Group

→ Ring
→ Abelian group under Addition
→ Monoid under Multi-plication

→ field
→ Abelian group under both Addition & multiplication.
→ Additive inverse, multiplicative inverse

Sup Lagraye's Theorem: O(H) [order of subgroup] is divisor of O(G).
↳ Apply (a,b) should belongs to H. → for subgroup verification.

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→ Union closed ✗
→ Intersection closed ✓ ⇏ for subgroup.

Cyclic Group:

If using an ele 'e', we are able to generate all other elements.

→ If a is generator then a^{-1} is also a generator.

No of Generators:

$G = \phi(n)$
↳ Relative primes

$\phi(7) = \{1,2,3,4,5,6\} = 6.$
$\phi(35) = \phi(7\times5) = \phi(7) \times \phi(5)$
$\phi(4) = \phi(2^2) = 2^2 - 2^1$

Functions:

A[m] B[n]

Total	1:1	onto
n^m	$m\times(m-1)\times(m-2)\times\cdots(m-n+1)$ ${}^mP_n$.	$\lvert A\rvert=\lvert B\rvert$ then $n!$ $\lvert A\rvert\neq\lvert B\rvert \Rightarrow \lvert A\rvert \geq \lvert B\rvert$ $n^m - {}^nC_1(n-1)^m + {}^nC_2(n-2)^m - {}^nC_3(n-3)^m + {}^nC_4(n-4)^m - \cdots$

Imp → If F and G both are one to one, then fog is also 1:1
→ If f and g both are onto, then fog is also onto.
→ If f and fog both are one to one then g is also 1:1.
Imp → If f and fog are onto, then g is not necessary onto.

$(fog)^{-1} = g^{-1}of^{-1}$

→ A fun is 1:1 iff either strictly increasing or decreasing.

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Combinatinoics

Que # of distinct positive factor / # divisor

eg. $2014 = 2^1 \times 19^1 \times 53^1$

$(1+1)(1+1)(1+1) = 8$ Ans

Que Dp. r balls in n box can be distributed as-

$(n-1+r C_r)$ Ans.

Imp Series:

$$[1+x+x^2+x^3+\cdots] = \frac{1}{(1-x)}$$

$$[1-x]^n = 1 - nx + \frac{n(n-1)}{2!}x^2 - {}^nC_3 x^3 + \cdots$$

$$[1-x]^n = 1 + (n)(-x) + \frac{n(n-1)}{2!}(-x)^2 + \cdots$$

Imp.

$$[1-x]^{+1} = 1 + (-1)(-x) + \frac{(-1)(-1-1)(-x)^2}{2!} + \cdots$$ Ans

$$[1-x]^{-n} = 1 + (-n)(-x) + \frac{(n)(n+1)}{2!}x^2 - \frac{n(n+1)(n+2)}{3!}x^3 + \cdots$$

Imp.

$$[1-x]^n = 1 + {}^nC_1(1)^1(-x) + {}^nC_2(-x)^2 + \cdots \quad \rightarrow \frac{n(n-1)}{2}$$

$$[1-x]^{-n} = 1 + {}^nC_1(x) + \frac{n(n+1)}{2}(-x)^2 + \frac{n(n+1)(n+2)}{2}(-x)^3 + \cdots$$

$$[1+x]^n = 1 + {}^nC_1 - n(x) + \frac{n(n-1)}{2!}(x)^2 + \frac{n(n-1)(n-2)}{3!}(x)^3 + \cdots$$

$$[1+x]^{-n} = 1 - n(x) + \frac{n(n+1)}{2!}x(x)^2 - \frac{n(n+1)(n+2)}{3!}x(x)^3 + \cdots$$

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$$(1-x)^n = (-x+1)^n = {}^nC_0(-x)^0(1)^n + {}^nC_1(-x)^1(1)^{n-1} + {}^nC_2(-x)^2 + \dots$$

$$(1-x)^{-n} = (-x+1)^n = \sum_{k=0}^{\infty} C(n+k-1)(-1)^k(-x)^k \quad \text{++}$$

or

$$\sum {}^{n+r-1}C_r(-x)^r(1)^{n-r} \quad \text{++}$$

Que $a_1 = 6$, $a_n = 6n^2 + 2n + a_{n-1}$ then $a_{99} = ?$

$$a_n = 6n^2 + 2n + 6(n-1)^2 + 2(n-1) + a_{n-2}$$
$$a_n = 6n^2 + 2n + 6(n-1)^2 + 2(n-1) + 6(n-2)^2 + 2(n-2) + \dots + 6\cdot1 + 2\cdot1$$

$$6[n^2 + (n-1)^2 + (n-2)^2 + \dots + 1] + 2[n + (n-1) + (n-2) + \dots + 1]$$

Ap.

Que $\{a_n\}_{n=0}^{\infty}$ is $\frac{1+z}{(1-z)^3}$ then $a_3 - a_0 = ?$

$$(1+z)(1-z)^{-3} \rightarrow {}^{n+1-1}C_r$$
$$(1+z)\left[1 - {}^3C_1(-z)^1 + {}^4C_2(-z)^2 - {}^5C_3(-z)^3 + \dots\right]$$
$$(1+z)\left[1 + 3z + 6z^2 + 10z^3 + \dots\right]$$

$a_3 = (z=3) \Rightarrow 10 + 6$

$a_0 = (z=0) \Rightarrow 1_0$ $\Rightarrow a_3 - a_0 = 16 - 1 = 15$

Que the exponent of 11 in the prime factorization of 300! is -

$1 \times 2 \times 3 \times \dots \times 300$

$$\left[\frac{300}{11}\right] = 27_0 + 2$$

(121, 242 will include twice).

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Graph Theory

Vertexe

⇒ # Simple Graphs = ? ${}^{n}C_2$ ⇒ # Edges

↳ $2^{{}^{n}C_2}$ [two possibility. edges may present or not].

⇒ $\sum d(v) = 2|E(G)|$. [No of odd degree vertices are even].

⇒ Havel Hakimi Theorem: check simple graph is possible with degree seq or not.

$$\delta \leq \frac{2|E|}{|V|} \leq \Delta$$

δ → min degree, Δ → max degree, (*)

→ SubGraph:
↳ Delete either edges or vertices

→ Induced subgraph
→ only by deleting vertices, not edges.

→ Clique: Set of pairwise Adjacent Vertex.

→ Independent set: set of pairwise non-Adjacent Vertex.

→ Bipartite -  ⇒ $n^2/4$ ⇒ No odd length cycle

→ Regular - All have same degree.

→ Hypercube - k-dim cube is 2^k. # Edges $= k \cdot 2^{k-1} = \frac{2^k * k}{2}$

→ Independent Numbers.

✓ Vertex Cover: Vertex set cover all Edges
✓ Edge Cover: Edge set covers all Vertices.
✓ Matching: All non-Adjacent Edges
✓ Independent set: All non-adjacent vertices

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- Distance : Min path b/w u & v
- Diameter : Max^m distance for all u, v
- Eccentricity : of a vertex, $\max_{v \in V(G)} d(u,v)$
- Radius : → Min^m of eccentricity
- Centre : → All vertices whose eccentricity is Min^m

- Walk : Both Vertex & Edges can Repeat
- Trails : Edge can't Repeat
- Circuit : → closed Trail
- Path : Vertex & Edges both can't Repeat.
- cycle : → closed Path.

⇒ Matching : Edges no Common Vertex.

Perfect matching in $K_{m,n}$ (Bipartite Graph) is ⇒ Complete (Equal partition) ⇒ $n!$

PM in Complete Graph → Odd No → No Matching
→ Even No → K_{2n} → $\frac{(2n)!}{n! \times 2^n} = (2n-1)(2n-3)(2n-5)$ ... 1

- Vertex Cover : Using vertices Cover Edges.

$$|\text{Maximal Matching}| \leq |\text{Minimal Vertex Cover}| \leq 2 \cdot |\text{Maximal Matching}|$$

finding Min^m Vertex Cover is NP-Complete problem.

→ Min^m Vertex Cover in $K_{m,n} = \min(m,n)$

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Planarity:

→ Euler formula.

$\Rightarrow n - e + f = k + 1$ → K-Compound

when $k = 1$, $n - e + f = 2$

$\Rightarrow \sum_{i=1}^{n} \deg(f_i) = 2E$.

Sup.

K_5 & $K_{3,3}$ is not Planner.

Sup 1: for Atleast 3 vertices,

$3f \leq 2e \Rightarrow f \leq \frac{2e}{3}$ & $e \geq \frac{3f}{2}$

$n - e + f = 2 \Rightarrow e \leq 3n - 6$

& $f \leq 2n - 4$

→ Every Planner is 4-~~clo~~ Colorable. ✓✓

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Hamiltonian Circuit:

→ Dirac's theorem

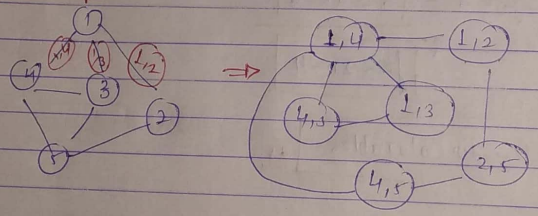
→ If $n \geq 3$, & each vertex $\deg(v) \geq n/2$ then G is Hamiltonian Cycle

→ $\deg(u) + \deg(v) \geq n$. → for every pair

↓

Ore's theorem.

Line Graph:



→ The line Graph of a cycle is cycle.

→ of Tree is not Tree.

→ of clique is not clique.

→ of Planne is not Planner.

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Graph Theory

Derrangement:

↳ changing position, None has fixed position.

Eg. 1 2 3 4 5 → Original

2 1 5 3 4 → D1

2 5 1 3 4 → D2

→ Compare with Original, Not with D_1, D_2 etc.

No position should match.

Eg. 1 2 3 4 5.

S_1 | 1 X X X X |

S_2 | X 2 X X X |

S_3 | X X 3 X X |

S_4 | X X X 4 X |

S_5 | X X X X 5 |

→ All these are Non-derayements.

$|S_1 \cup S_2 \cup S_3 \cup S_4 \cup S_5| \Rightarrow$

$S_1 = L4, S_2 = L4, S_3 = L4$ —

$|S_1 \cap S_2| = |S_2 \cap S_3| = |S_3 \cap S_4| = |S_4 \cap S_5| = L3$.

$|S_1 \cap S_2 \cap S_3| = _ _ = L2$

$|S_1 \cap S_2 \cap S_3 \cap S_4| = _ _ = L1$

$|S_1 \cap S_2 \cap S_3 \cap __| = L0 = 1$.

So, $\sum |S_i| - \sum |S_i \cap S_j| + \sum |S_i \cap S_j \cap S_k| - _ _ _$

$\Rightarrow {}^{5}5 * 4! - {}^{5}C_2 * L3 + {}^{5}C_3 * L2 - {}^{5}C_4 * L1 + {}^{5}C_5 * L0$ (+)

↓ Non-derayements. ↓ 76.

$\Rightarrow$ # Derayements = total − (Non-derayements)

L5 − 76

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